Computer Science Coursework

Discussion

Introduction to and Overview of the Business

The Mapleview Private Healthcare Group has a hospital located in Antrim, by the name of Mapleview Hospital. This hospital, alongside providing emergency care, provides specialist services, including diagnostics, outpatient clinics, inpatient care and day-case treatments for both medical and surgical needs. The hospital hires and works with a range of employees: there are Senior Nurses, Nurses, Healthcare Assistants, and Nursing Assistants, alongside working with Private Doctors and Nurses and multiple secondary and supporting roles.



Currently the hospital incorporates a hybrid system for managing patients. They have an Electronic Health Record (EHR), for digitally keeping track of patients, but they also use paper systems, due to some limitations of the digitised system. I have been contacted by a Senior Nurse to explore possible system replacement ideas with them. Currently, the Nurses have general patient information on a physical document, and they have detailed information on each patient on their digital system.

Many of the Nurses in the ward are advocating for an all-digital system replacement, as accidents and mix-ups can occur with paperwork and documents. They are also keen on new features and capabilities being added to the new system, such as diet tracking, and automated reminders. A few of the Healthcare Assistants have come to me and asked for an automated medicine label maker, that will make a printable pdf file for medicine that is to be taken by patients. Doctors have also voiced their opinions on communication between them and the Nurses, stating that it would be a useful feature to add a comment line in the digital files to allow for some explanations of certain tasks.

The Senior Nurses have decided to go through with these ideas, and have added one of their own:

- The Senior Nurses would like an easy way to view every patient in their ward.
- The new system will need to include everything that the old system had, alongside the new additions.
- The system will include timed reminders with recurrence built in.
- All features should be built with shift swapping in mind.
- There will be an emphasis placed on clear communication between healthcare workers.

Due to the nature of the system, the proposed replacement will need to be robust and efficient. This is because any systems crashes or bugs in the system will hinder the healthcare workers and possibly put lives at risk.

The system will have a user base of: Senior Nurses, Nurses, Healthcare Assistants, and Nursing Assistants, alongside working with Private Doctors and Nurses and multiple secondary and supporting roles.

Each of these will need its own access level and granted permissions.

The computer-based proposed system should allow for the Nurses and Senior Nurses to:

- Enter and update general information on their patients.
- Enter and update medical information on their patients.
- Enter medicine required by their patients and print off labels.
- Communicate with other healthcare workers using comments.
- Create treatment plans for patients with set reminders that can repeat.
- Log and update vital signs of the patient.
- Update and review information being monitored on the patient, with reminders to do so.
- Create, update, and review patient care plans.
- Discharge patients.
- Communicate directly in a communication tab that stores messages in a file format.
- Coordinate handoffs.
- Calculate free beds and free spots *for* beds in the ward.
- Create a PDF with a summary of treatments.
- View graphs created for monitoring information or vital signs against time.
- Enter or update contact information for relatives.
- Calculate medicine stock (Senior only).

The system should allow for the Healthcare Assistants and Nursing Assistants to:

- Update medical information on patients.
- Enter dietary tracking information on patients.
- Enter treatments and get them printed off to be signed by a Nurse.
- Communicate using comments.
- Read and complete reminders from treatment plans.
- Update vital signs of the patient.
- Update and review information being monitored on the patient, with reminders to do so.
- Review patient care plans.
- Communicate in the communication tab.
- View graphs created for monitoring information or vital signs against time.
- Create a PDF with a summary of treatments.

The system should allow for the Doctors to:

- Access and update general information on their patient.
- Access and update medical information on their patient.
- Enter treatments for their patient and print them off to be signed and administered by a Nurse.
- Read patient's dietary tracking.
- Communicate using comments.
- Create treatment plans that set reminders that can repeat.
- Review information being monitored on the patient.
- Review vital signs of the patient.
- Update and review patient care plans.
- Communicate directly in a communication tab that stores messages in a file format.
- Create a PDF with a summary of treatments.

Contact details can be set up for relatives of patients, but only with help from the Nurses, so, on the Nurses computers.

[might also want to add email-based limitations]

As with all systems, there are limitations:

- I'd like the system to be connected online. The communication part of the system would be very useful if live, but unfortunately not something that I'd be able to implement right now.
- The system does not have a patient front-end. It would be helpful for patients to see a very brief summary of their file on system.
- It would be very helpful to integrate the system with an existing system the Nurses use to administer medicine. The Nurses and meant to scan a wristband on a patient before administering medicine to them, to make sure they are treating the right patient. However, it would be difficult for me to introduce a camera tool and QR code reader into the system.
- The Healthcare Assistants will need to manually enter vitals and update information being monitored, because an API couldn't be set up to interact with the medical devices and automatically ping them for the information to be updated.